The Derailing of the MBTA

Johnny Scaramuzzo

Problem Statement

- Is the MBTA providing reliable, predictable, and satisfactory service across lines and time periods?

Motivation

- As a resident of a Boston suburb, I frequently find myself using the MBTA as a means to avoid traffic going into the city.
- However, there have been many times where I have encountered difficulties in trying to get in and out of Boston efficiently.
- I decided to study how reliable the MBTA is as a whole.

- Since the midterm, this problem statement has remained the same, and I became even more interested in its causes and solutions when I was unsatisfied with the MBTA service over Thanksgiving break.

Data Source Overview

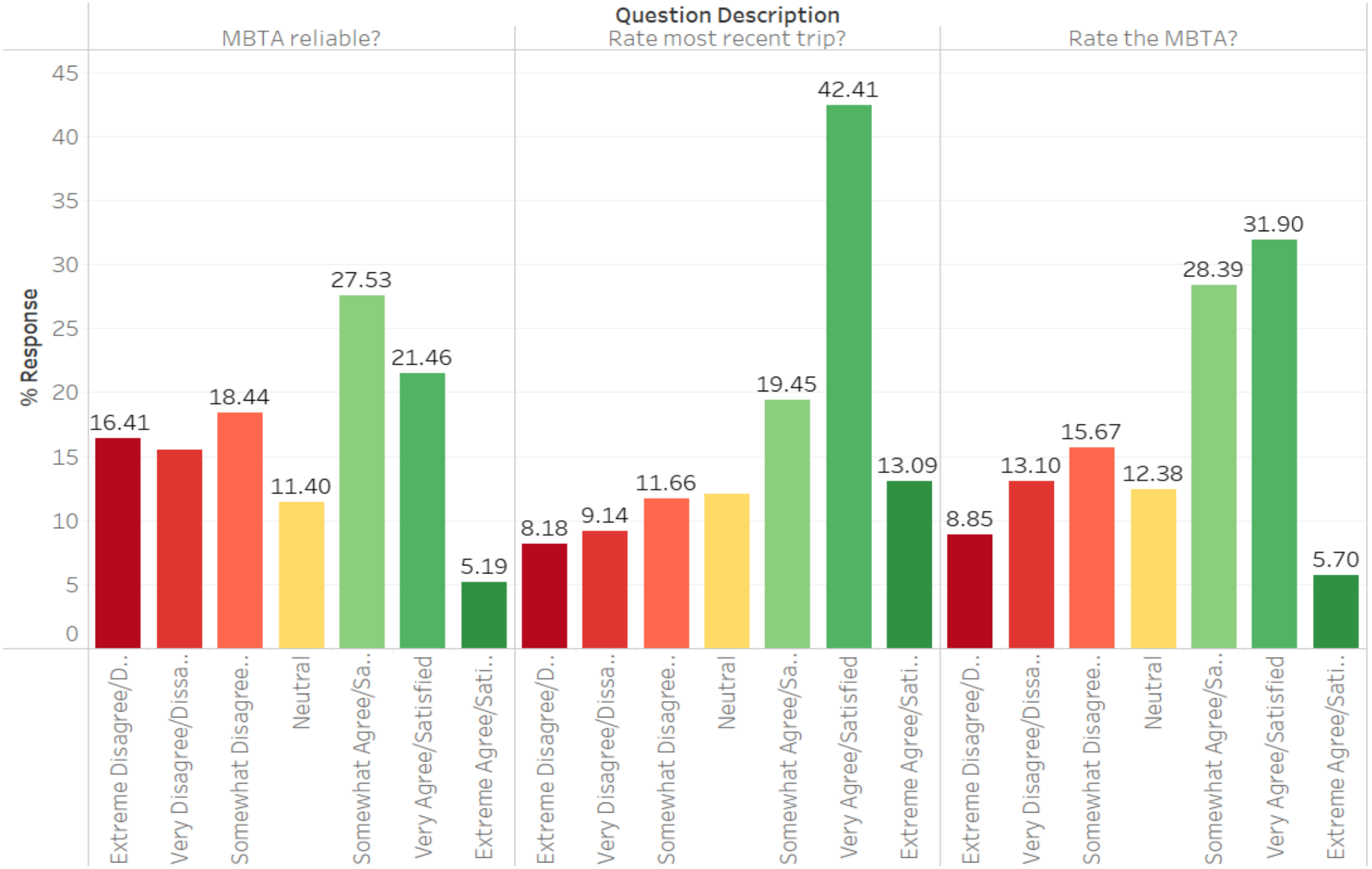
- 4 data sets obtained from MassDOT public records (2024-25):
  - Service Alerts
    - Information regarding disruptions to normal MBTA services. Contained reason for service, mode of transportation, and severity score.
  - Expected vs Actual Arrival Times
    - The predicted times for MBTA arrivals compared to the actual time of arrival. Separated by mode of transportation, and contained details such as time of day and ridership.
  - Rider Satisfaction Survey
    - An anonymous survey sent out to riders after their trip on the MBTA, asking about their experience on an “Extremely Dissatisfied” to “Extremely Satisfied” scale
  - Work Orders
    - Pre-processed and cleaned with help of Professor. Contained information on work orders called by the MBTA such as the length of time and cost to complete them.

Introduction

MBTA Reliability

- Results of an anonymous post-trip survey show that while the MBTA is viewed favorably as a whole, major questions are raised regarding its reliability:

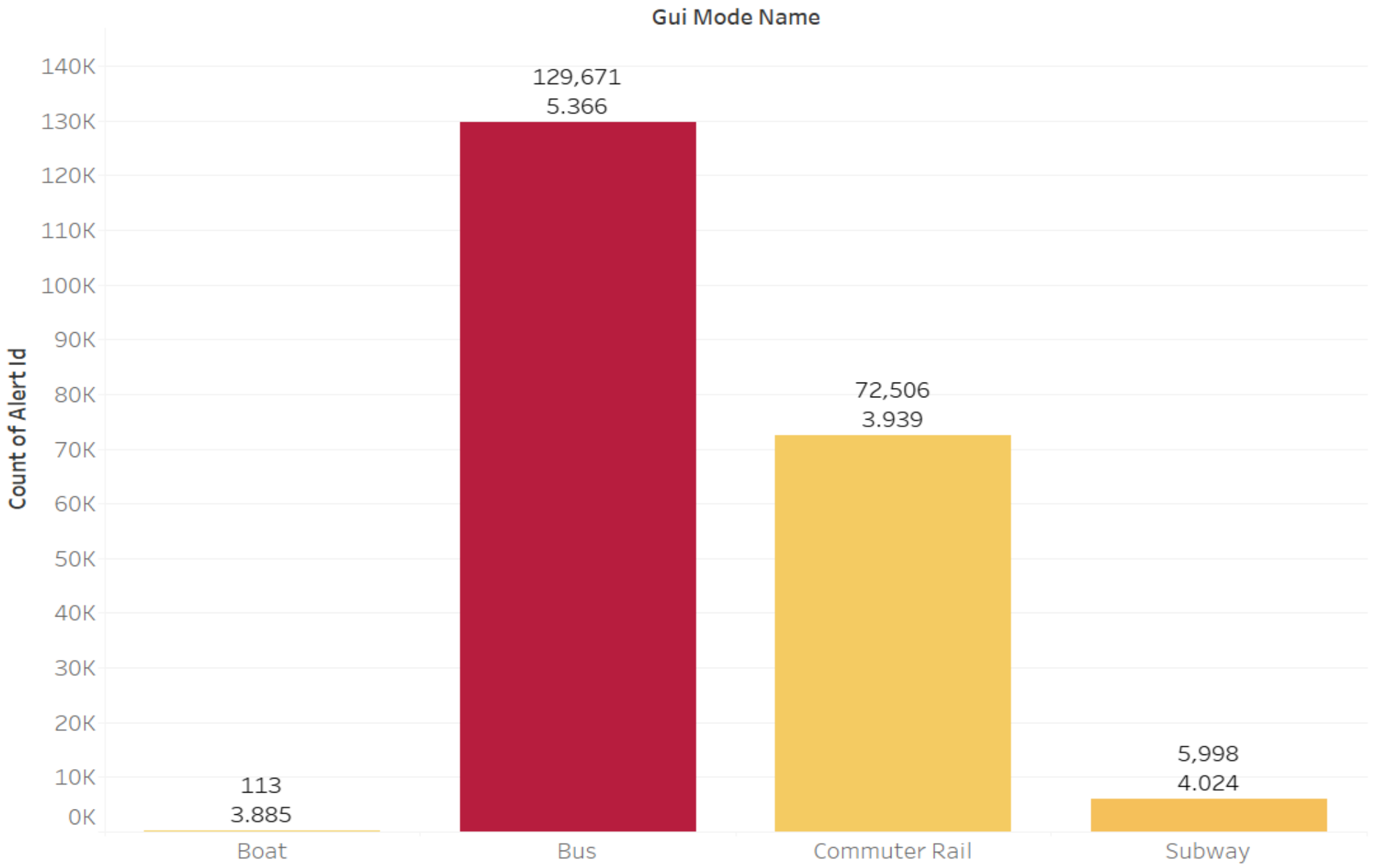
Satisfaction



Measure Names

- Extreme Disagree/Dis..
- Very Disagree/Dissati..
- Somewhat Disagree/D..
- Neutral
- Somewhat Agree/Sati..
- Very Agree/Satisfied
- Extreme Agree/Satisfi..

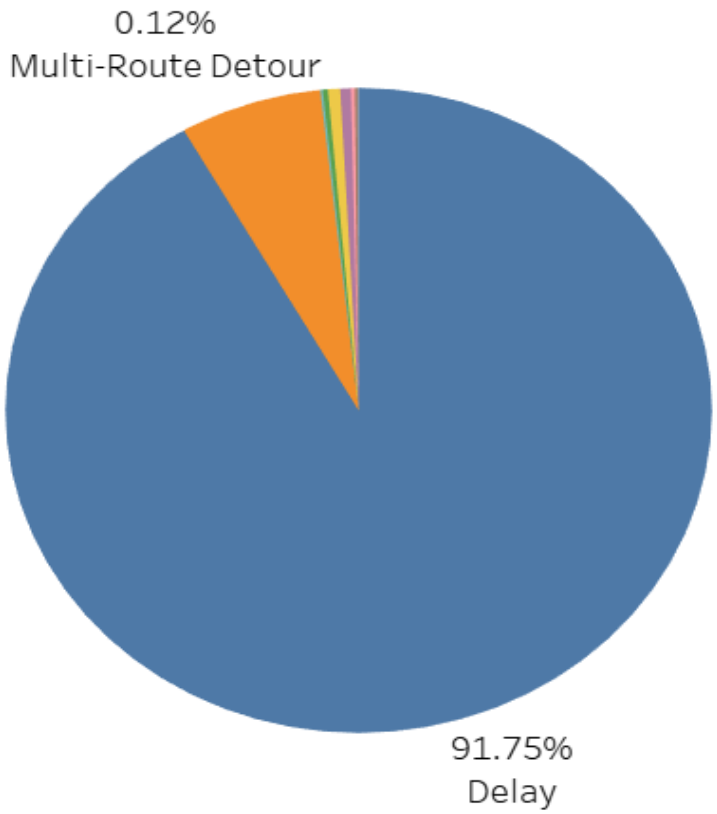
Service Alerts by Mode and Average Severity Score



AVG(Severity Code)

3.885 5.366

Buses



Effect Name

- Delay
- Detour
- Extra Service
- Multi-Route Detour
- Schedule Change
- Service Change
- Snow Route
- Stop Closure
- Stop Move
- Suspension

CNT(Effect Name)

129,671

Bus Delay Causes

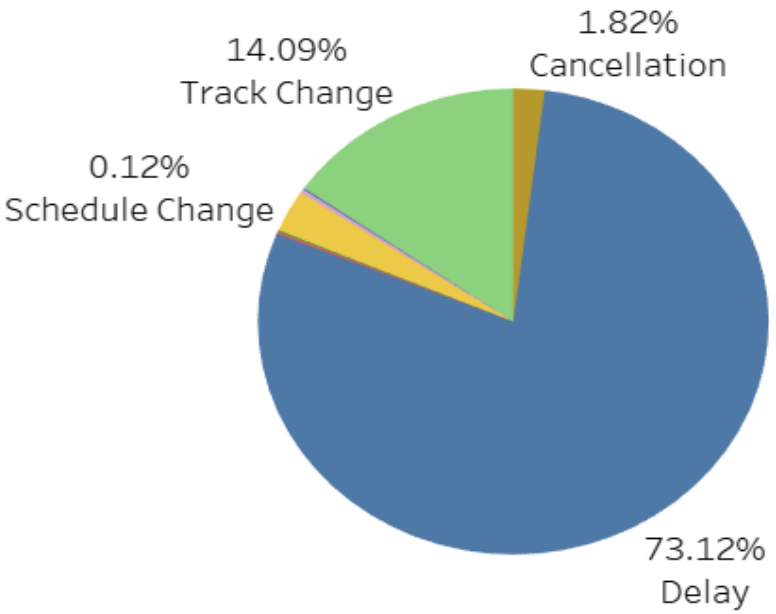
CNT(Cause Name)

6 72,009

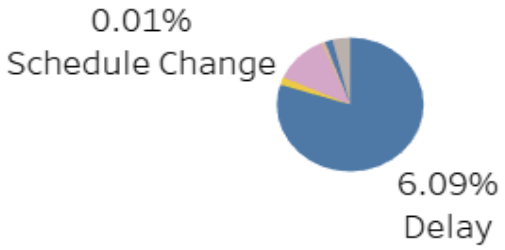
Rail

Gui Mode Name

Commuter Rail



Subway



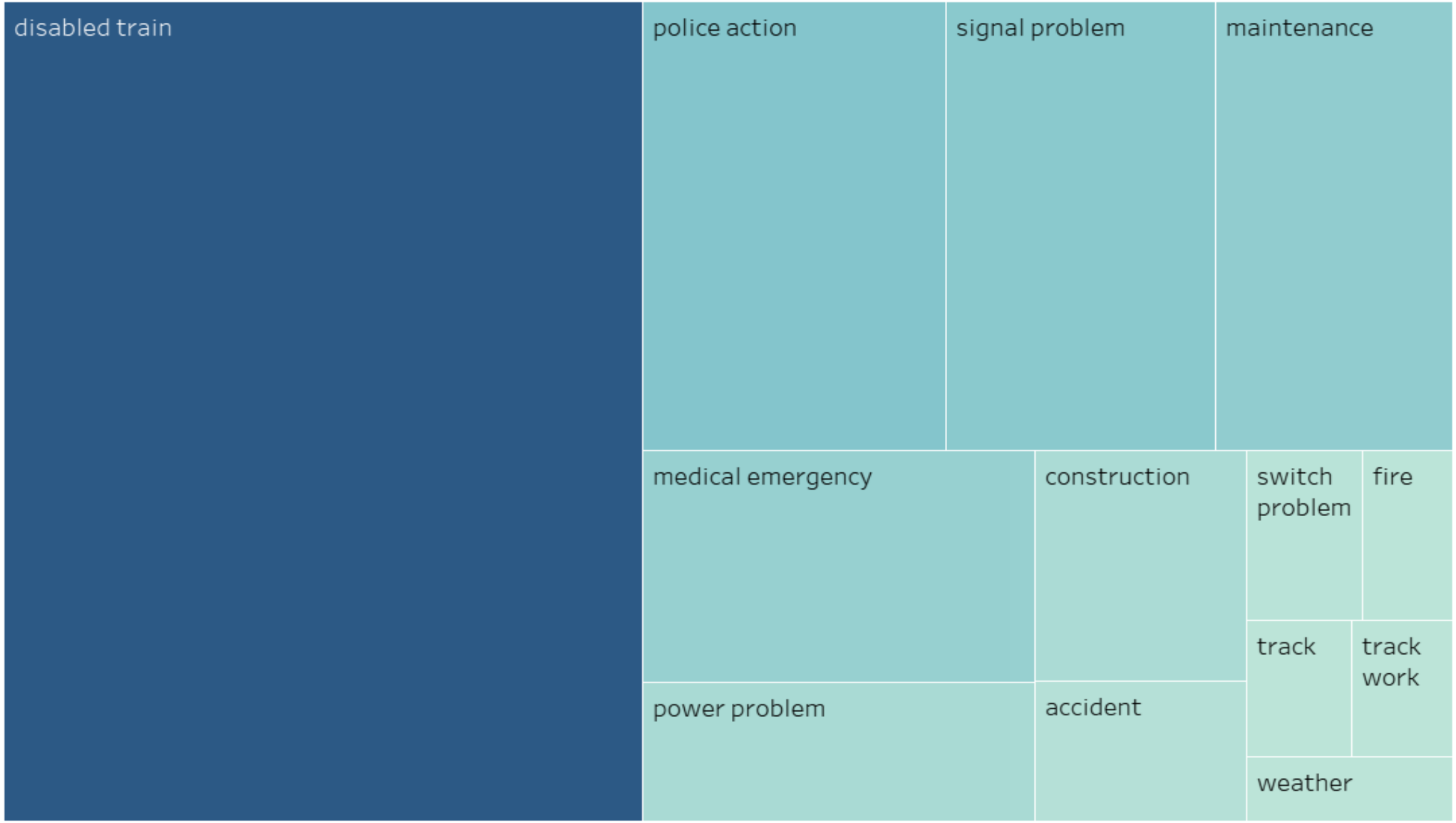
CNT(Effect Name)

5,998
20,000
40,000
60,000
72,506

Effect Name

- Cancellation
- Delay
- Extra Service
- Schedule Change
- Service Change
- Shuttle
- Station Closure
- Station Issue
- Suspension
- Track Change

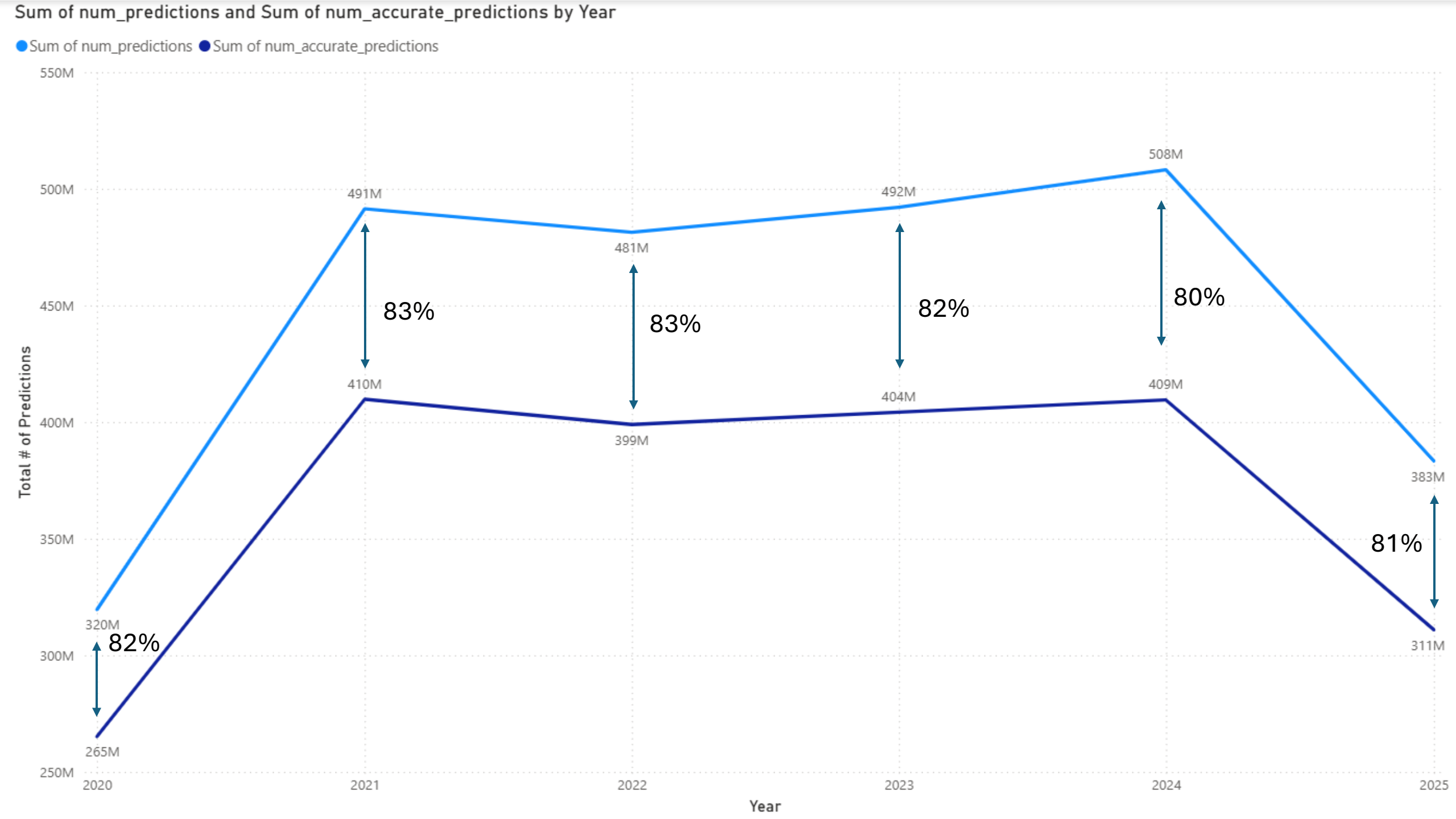
Rail Delay Causes



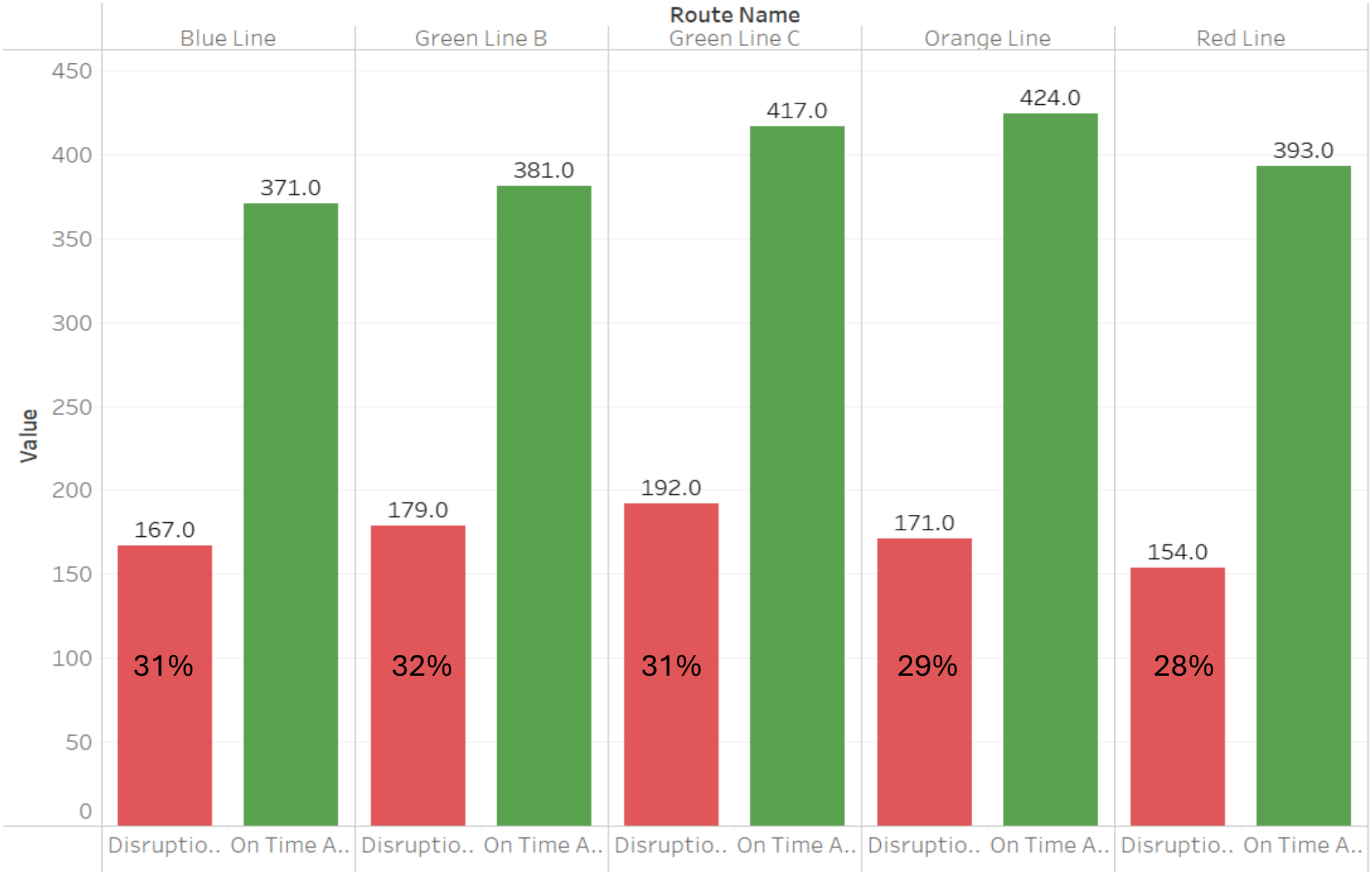
CNT(Cause Name)

44 1,723

Core Evidence


Sum of num_predictions and Sum of num_accurate_predictions by Year
Sum of num_predictions
Sum of num_accurate_predictions
550M
500M
450M
400M
350M
300M
250M
Total # of Predictions
2020
2021
2022
2023
2024
2025
Year
320M
491M
481M
492M
508M
383M
265M
410M
399M
404M
409M
311M
82%
83%
83%
82%
80%
81%

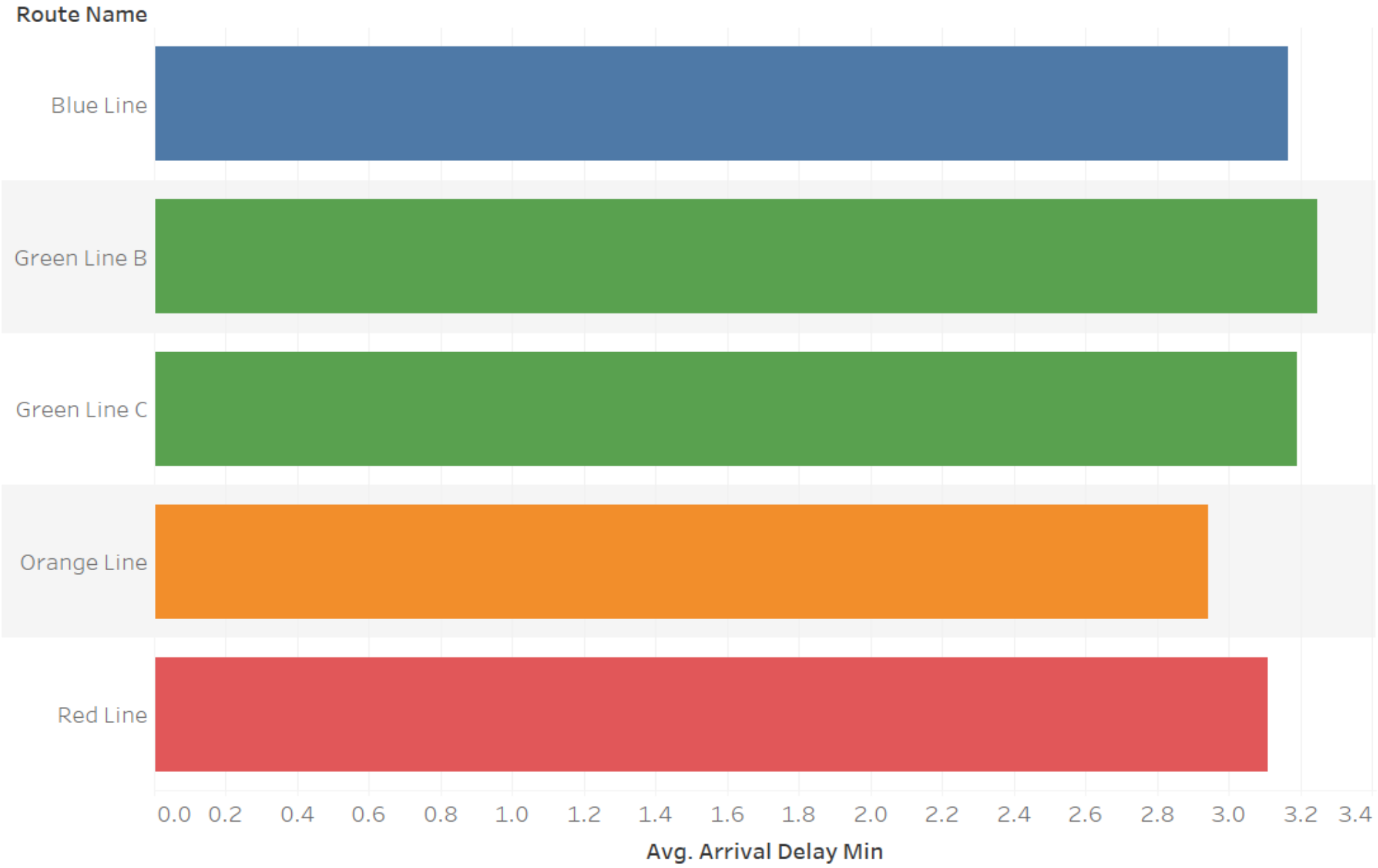
On Time vs Disruption



Measure Names

- Disruption Flag
- On Time Arrival Flag

Delay by Line



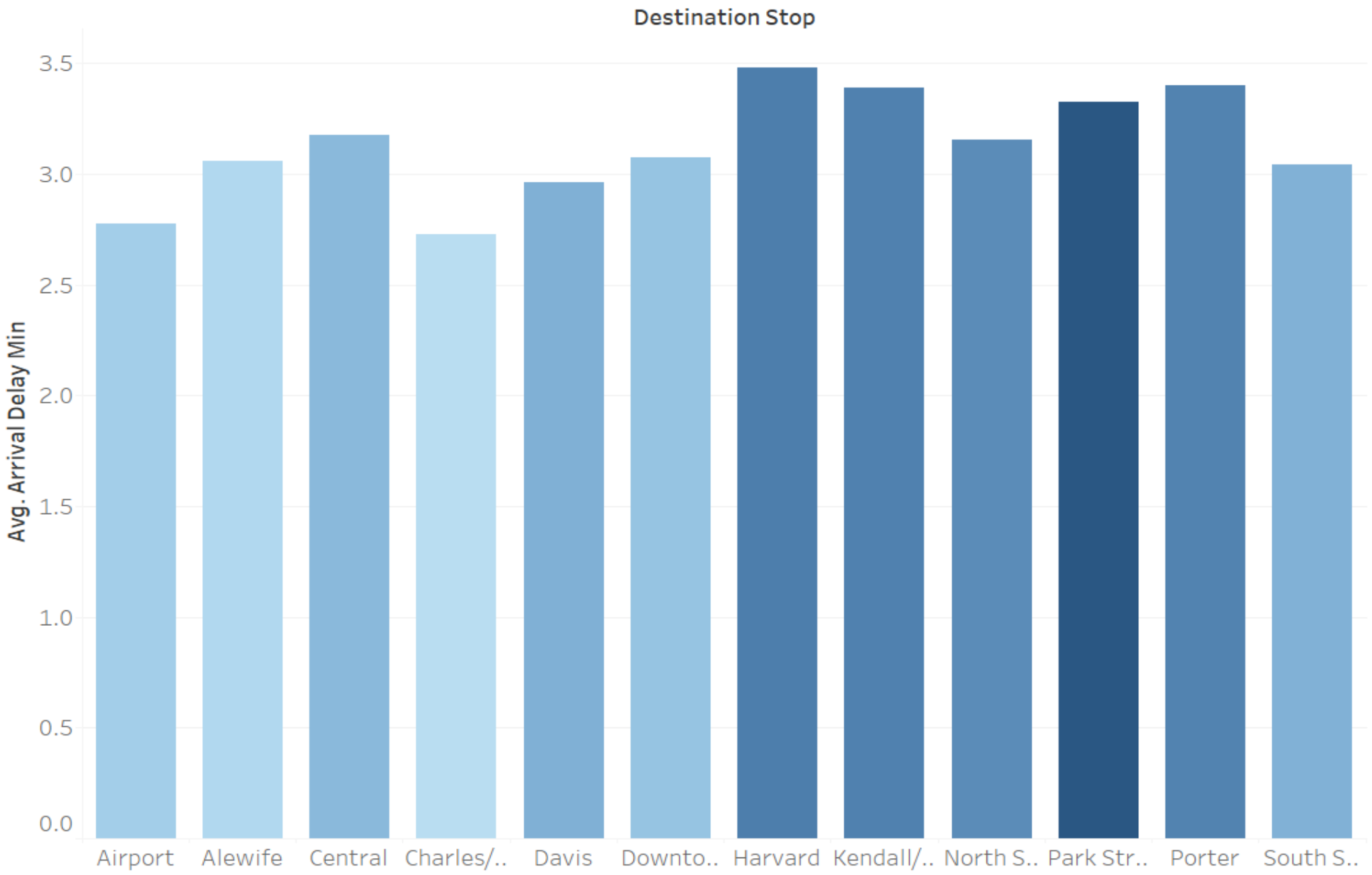
Route Name

- Blue Line
- Green Line B
- Green Line C
- Orange Line
- Red Line

Highlight Route Name

Green Line B ×

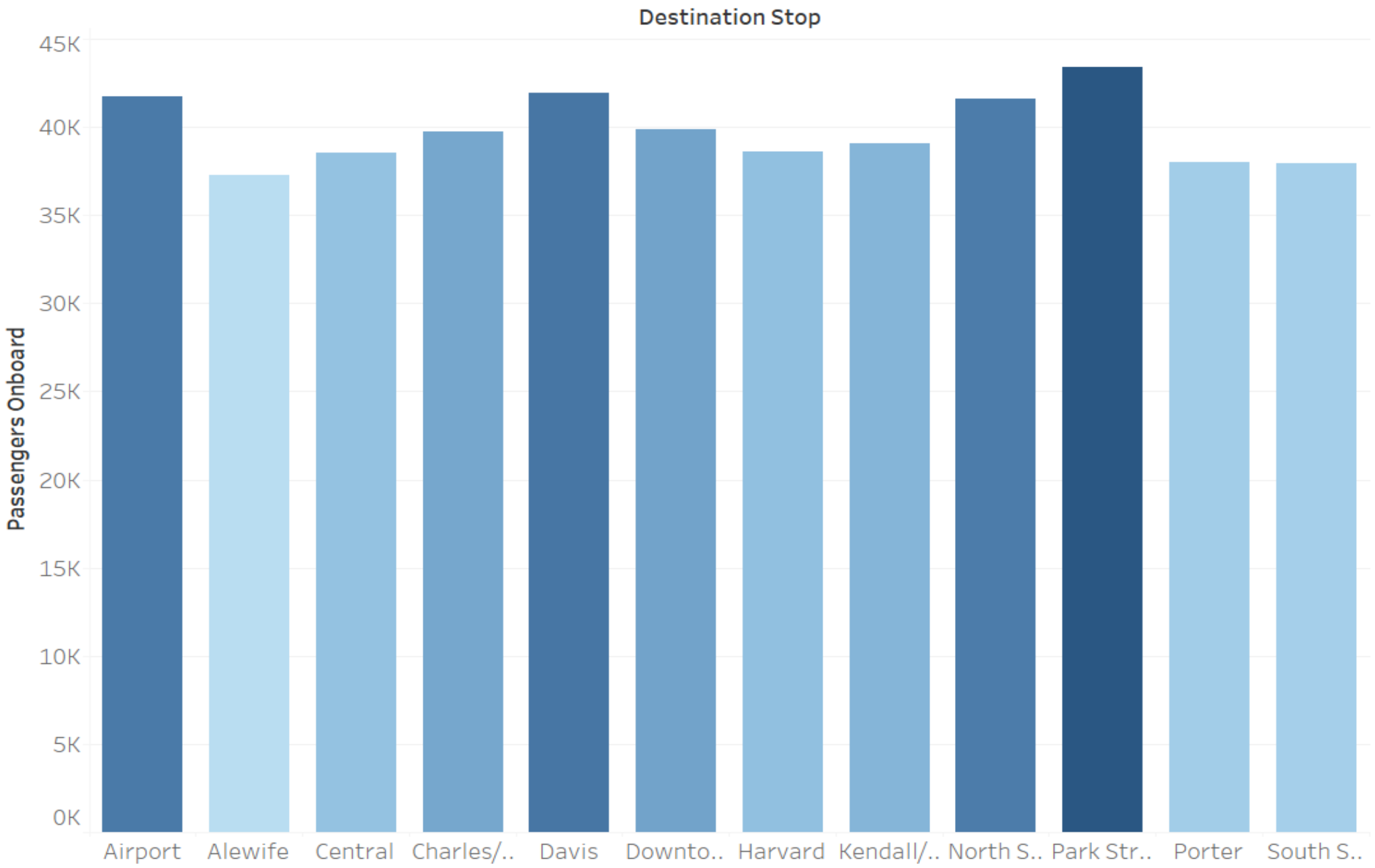
Delay by Stop



SUM(Arrival Delay Min)

829.0 1,094.2

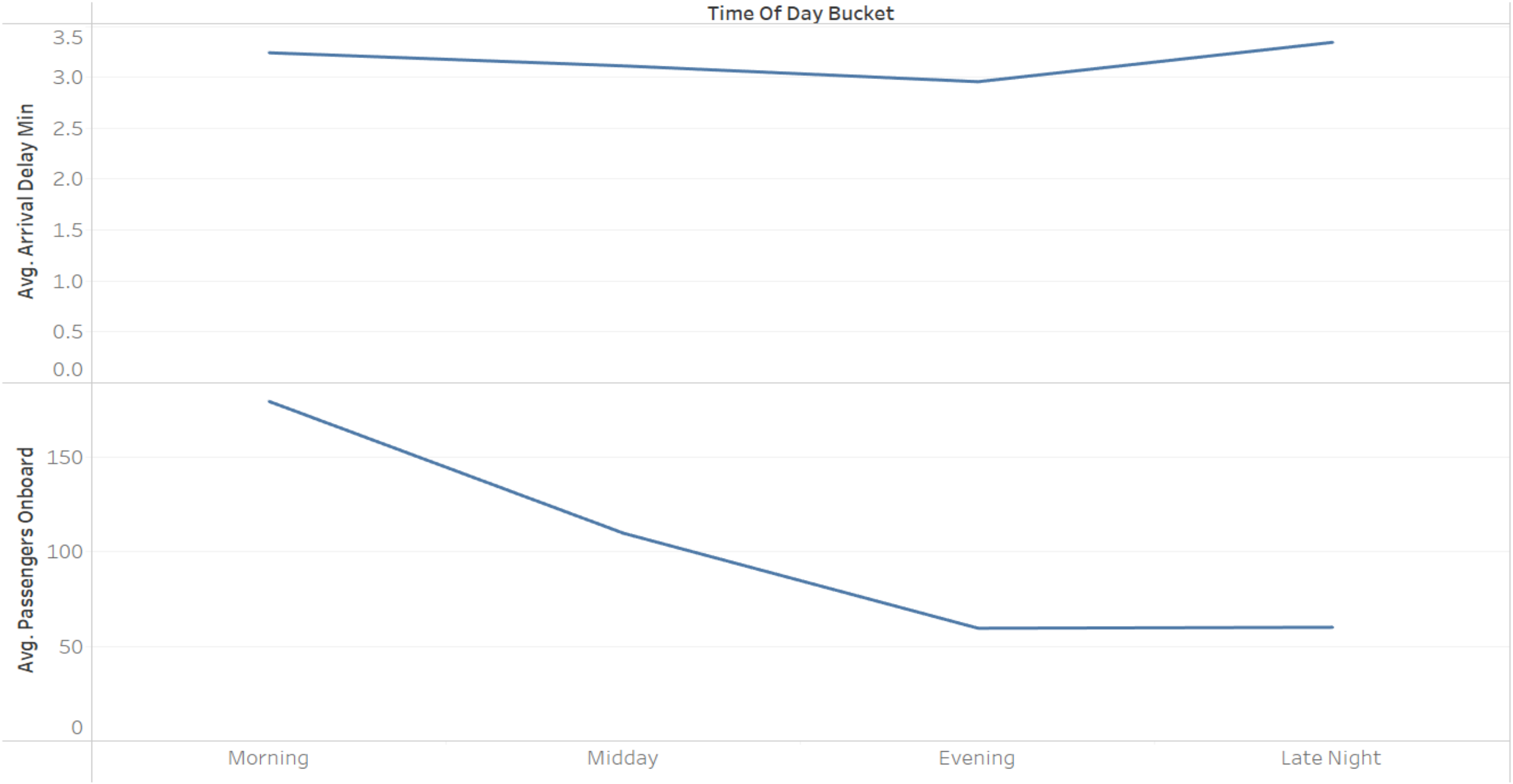
Stop by Ridership



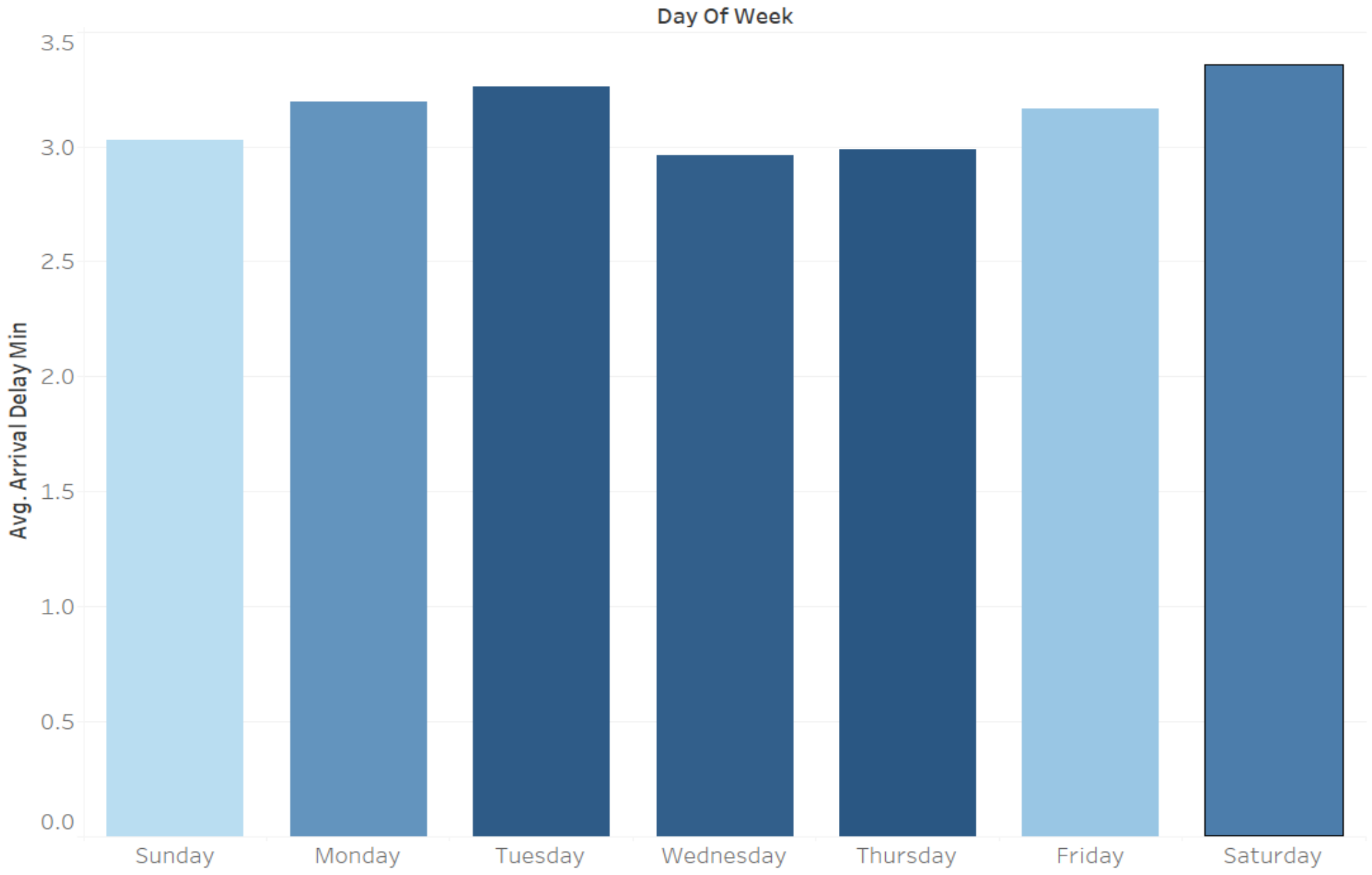
SUM(Passengers Onbo...

37,278 43,433

Delay Trends



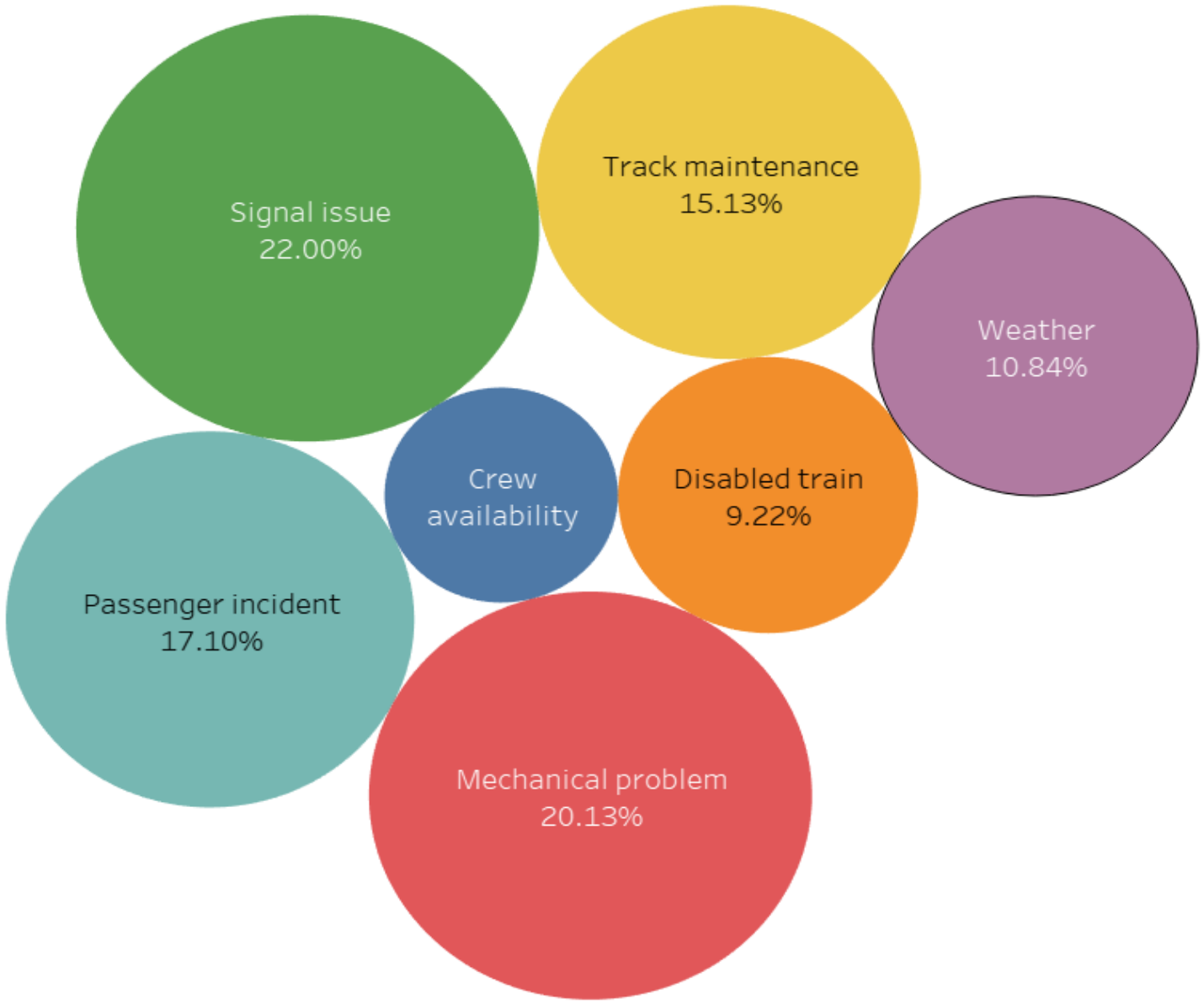
Delay by Day



SUM(Arrival Delay Min)

1,327.3 1,776.9

% of Delay Mins Accounted For

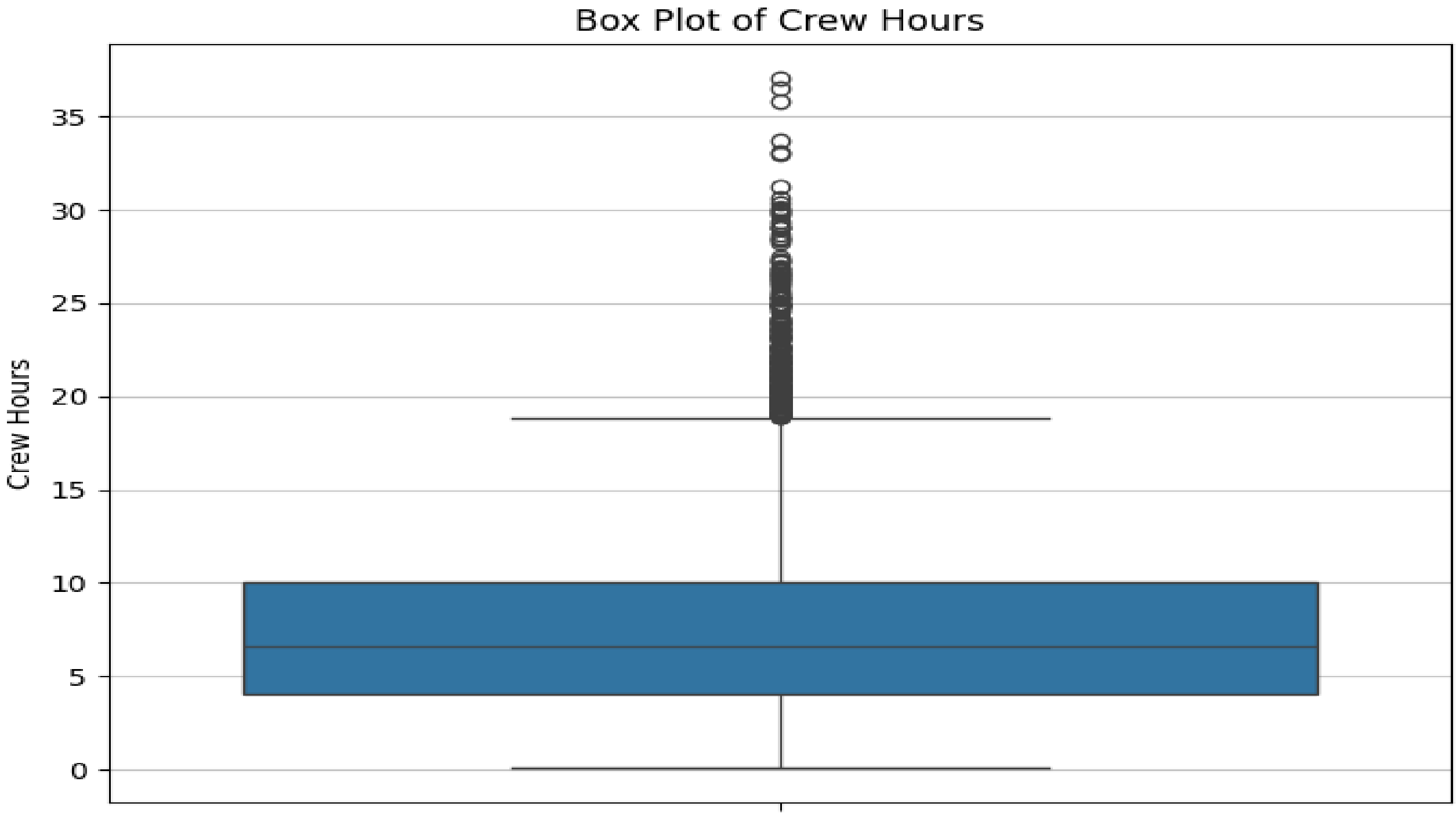


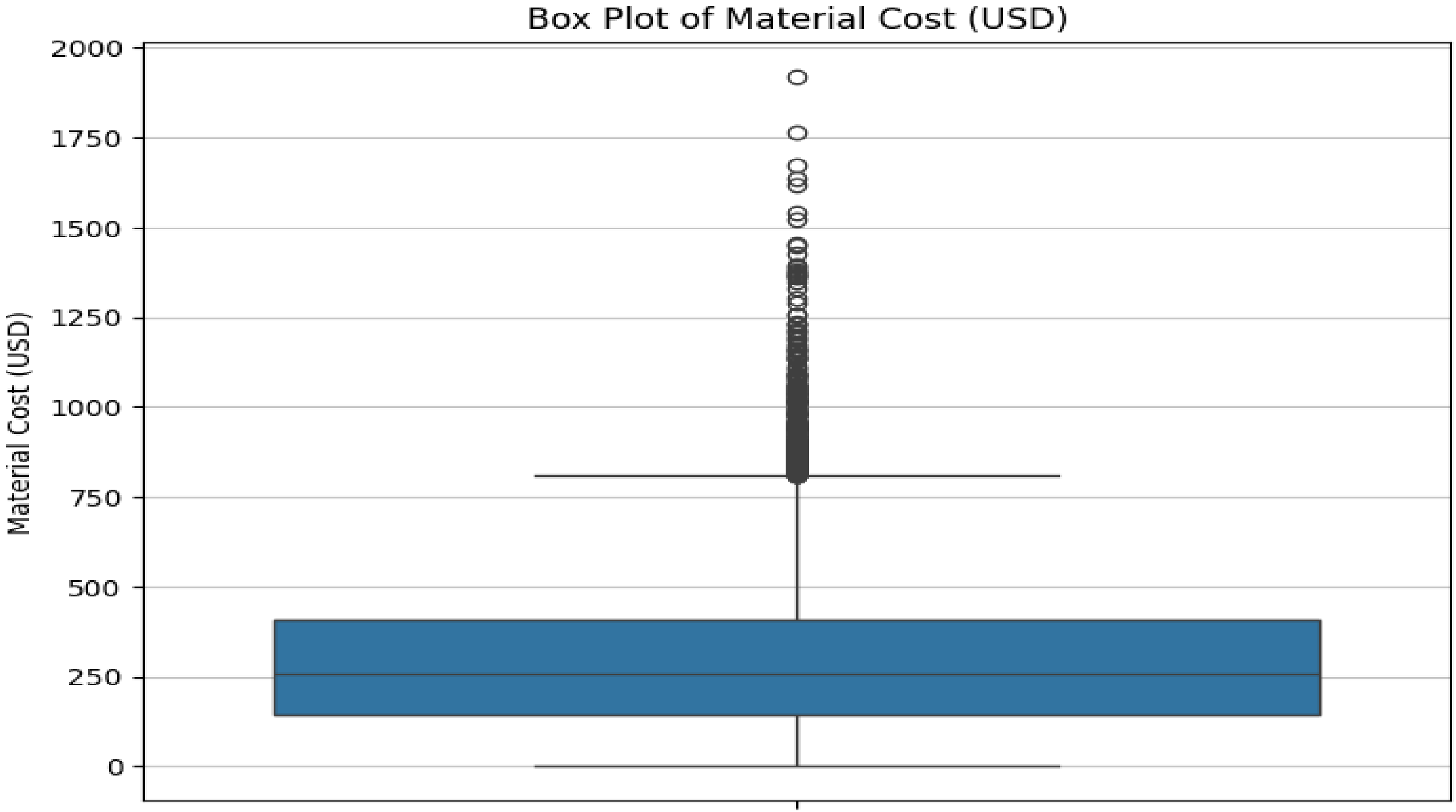
Primary Disruption Rea...

- Crew availability
- Disabled train
- Mechanical problem
- Passenger incident
- Signal issue
- Track maintenance
- Weather

```python
import pandas as pd
import numpy as np
df = pd.read_csv('/content/work_orders.csv')
print(f"Avg hours worked = {df['crew_hours'].mean():,.2f}")
print(f"Max hours worked = {df['crew_hours'].max():,.2f}")
print(f"Total hours worked = {df['crew_hours'].sum():,.2f}")
print(f"Avg cost of job = {df['material_cost_usd'].mean():,.2f}")
print(f"Max cost of job = {df['material_cost_usd'].max():,.2f}")
print(f"Total cost of jobs = {df['material_cost_usd'].sum():,.2f}")
```

```
Avg hours worked = 7.57
Max hours worked = 37.00
Total hours worked = 75,651.40
Avg cost of job = 303.42
Max cost of job = 1,919.27
Total cost of jobs = 3,034,185.66
```


Box Plot of Crew Hours
Crew Hours
35
30
25
20
15
10
5
0


Box Plot of Material Cost (USD)
Material Cost (USD)
2000
1750
1500
1250
1000
750
500
250
0

Resolution

Potential Fixes

Modernize Rail Infrastructure: Over 42% of delay minutes are caused by signal issues and mechanical problems, therefore the MBTA should prioritize investments in upgrading signal systems and maintenance. These are internal failures under the MBTA's control, unlike traffic.

More Bus Lanes: The data proves that bus reliability is almost entirely compromised by traffic. To fix the high severity of bus alerts, the MBTA should create more dedicated bus lanes and separate their transit from general congestion around Boston.

Weekend Operations: The spike in delays on Saturdays suggests a lapse in weekend planning by the MBTA. Service and maintenance schedules should be adjusted to ensure weekend reliability matches weekday standards.

Predictive Maintenance: The extreme outliers in the work orders such as 37 hour shifts shows horrible equipment failures that cause long repair sessions. Moving to a predictive maintenance model could catch these issues earlier, turning emergency overtime repairs into routine and cheaper fixes.

References

- *“MBTA Blue Book Open Data Portal.”* **MBTA-MassDOT Open Data**, https://mbta-massdot.opendata.arcgis.com/search?tags=performance. Accessed *[12/12/25]*.